SERKAN SULUN

DEEP LEARNING RESEARCHER

CONTACT

Porto, Portugal

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EDUCATION

DOCTORATE
Electrical and Computer Engineering
University of Porto
2019 - Present

MASTERS
Electrical and Electronics Engineering
Koc University
2016 - 2018
GPA: 3.81

BACHELORS
Electronics Engineering
Sabanci University
2009 - 2014
GPA: 3.86

SKILLS

Deep learning
Pytorch | Tensorflow | Caffe | Generative Networks | Adversarial Learning | Recurrent Neural Networks | Reinforcement learning

Programming
Python | MATLAB | C++ | Julia

Signal processing
Audio processing | Image processing | Video processing | Video compression | Computer vision

Music production
Cubase | Protools | Audacity | Music21 | MIDI

Miscellaneous
Remote development | Machine learning | Git | LaTeX | Ubuntu | Windows

LANGUAGES

- **English** | *Fluent*
- **Turkish** | *Fluent*
- **Portuguese** | *Beginner*
- **German** | *Beginner*

SUMMARY

I use deep neural networks on music and video.

EXPERIENCE

RESEARCHER

INESC TEC, Portugal | Mar 2019 - Present

Enhancement of complex music signals using deep neural networks

Video-based symbolic music generation

RESEARCH ASSISTANT

Koc University, Turkey | Sep 2016 - Nov 2018 (2 years 2 months)

Video compression and frame prediction using deep neural networks, including generative adversarial networks

TEACHING ASSISTANT

Koc University, Turkey | Sep 2016 - Nov 2018 (2 years 2 months)

Leading laboratory sessions and occasionally lectures on programming with MATLAB

SOUND ENGINEER

Telfs Youth Center Studio, Austria | Oct 2014 - Aug 2015 (11 months)

Recording and mixing music performed by local youngsters

INTERN

Champalimaud Neuroscience Program, Portugal | Jul 2013 - Oct 2013 (3 months)

Processing neural signals and videos obtained by rat experiments, designing the controller board. Article acknowledgement: ncbi.nlm.nih.gov/pmc/articles/PMC4721960

PEER INSTRUCTOR

Sabanci University, Turkey | Feb 2010 - Jun 2014 (4 years 3 months)

Lecturing in small study groups and occasionally in amphitheaters on freshman chemistry and biology

PUBLICATIONS

ON FILTER GENERALIZATION FOR MUSIC BANDWIDTH EXTENSION USING DEEP NEURAL NETWORKS (2020)

IEEE Journal of Selected Topics in Signal Processing

DOI:10.1109/JSTSP.2020.3037485 | arXiv:2011.07274

CAN LEARNED FRAME-PREDICTION COMPETE WITH BLOCK-MOTION COMPENSATION FOR VIDEO CODING? (2020)

Signal, Image and Video Processing

DOI:10.1007/s11760-020-01751-y | arXiv:2007.08922

DEEP LEARNED FRAME PREDICTION FOR VIDEO COMPRESSION (2018)

Koc University, Masters Thesis

Thesis:531218 | arXiv:1811.10946

AWARDS

- **Doctoral Fellowship** | la Caixa Foundation
- **Graduate Fellowship (Masters)** | Koc University
- **Ranked 9 out of 204,392 in Turkish Academic Personnel and Graduate Education Exam**
- **Excellence Merit Scholarship (Bachelors)** | Education Foundation of Turkey
- **Excellence Merit Scholarship (Bachelors)** | Sabanci University
- **Employment Scholarship** | Sabanci University
- **Bachelors Scholarship** | Prime Ministry of Turkey
- **Ranked 297 out of 1,349,782 in Turkish University Entrance Exam**
- **Erasmus+ Exchange Grant** | European Commission
- **Erasmus+ Internship Grant** | European Commission

INTERVIEWS

OBSERVADOR (2020)

Generating music from videos

Portuguese original | English translation

OLHAR NO PROGRESSO (2021)

Deep learning, multimedia and music

Newspaper website | YouTube